UNIT1
INTRODUCTION TO VIRTUALIZATION

1. Virtualization
Virtualization is technology that you can use to create virtual representations of servers, storage, networks, and other physical machines. Virtual software mimics the functions of physical hardware to run multiple virtual machines simultaneously on a single physical machine. Businesses use virtualization to use their hardware resources efficiently and get greater returns from their investment. Virtual machines and hypervisors are two important concepts in virtualization.

Virtual machine
A virtual machine is a software-defined computer that runs on a physical computer with a separate operating system and computing resources. The physical computer is called the host machine and virtual machines are guest machines. Multiple virtual machines can run on a single physical machine. Virtual machines are abstracted from the computer hardware by a hypervisor.

Hypervisor
The hypervisor is a software component that manages multiple virtual machines in a computer. It ensures that each virtual machine gets the allocated resources and does not interfere with the operation of other virtual machines.
There are two types of hypervisors.

Type 1 hypervisor
A type 1 hypervisor, or bare-metal hypervisor, is a hypervisor program installed directly on the computer's hardware instead of the operating system. Therefore, type 1 hypervisors have better performance and are commonly used by enterprise applications. KVM uses the type 1 hypervisor to host multiple virtual machines on the Linux operating system.

Type 2 hypervisor
Also known as a hosted hypervisor, the type 2 hypervisor is installed on an operating system. Type 2 hypervisors are suitable for end-user computing.

1.1 Types of Virtualization
1. **Hardware Virtualization (Server Virtualization)**
   - **Definition:** The process of creating virtual machines (VMs) that run on a physicalhost machine, each acting as an independent computer with its own operating system and applications.
   - **Key Technologies**:
     - **Hypervisors:** Software layers that enable multiple operating systems to share a single hardware host. Examples include:
       - **Type 1 (Bare-Metal) Hypervisors:** VMware ESXi, Microsoft Hyper-V, and KVM.
       - **Type 2 (Hosted) Hypervisors:** VMware Workstation, Oracle VM VirtualBox.
   - **Benefits:**
     - Improved resource utilization.
     - Isolation of applications.
     - Simplified management.

2. **Software Virtualization**
    - **Definition:** Allows applications to run in isolated environments, often using containers.
    - **Key Technologies:** Docker, Kubernetes, LXC (Linux Containers).
    - **Benefits:**
        - Consistent environments across development, testing, and production.
        - Rapid deployment and scaling.
        - Efficient use of system resources.
3. **Desktop Virtualization**
    - **Definition:** Technology that allows desktop environments to be hosted on a central server and accessed remotely by users.
    - **Key Technologies***:*
        - **VDI (Virtual Desktop Infrastructure):** Citrix Virtual Apps and Desktops, VMware Horizon.
    - **Benefits:**
        - Centralized management of desktops.
        - Enhanced security and data protection.
        - Flexibility for remote work.
4. **Network Virtualization**
    - **Definition:** The process of creating a virtualized network that operates independently of the physical network infrastructure.
    - **Key Technologies:** VLANs (Virtual Local Area Networks), SDN (Software-Defined Networking), NFV (Network Functions Virtualization).
    - **Benefits:**
        - Simplified network management.
        - Improved network flexibility and scalability.
        - Enhanced performance and security.
5. **Storage Virtualization**
    - **Definition:** The pooling of physical storage from multiple devices into a single,logical storage unit that is managed centrally.
    - **Key Technologies:** SAN (Storage Area Network), NAS (Network Attached Storage), Storage Virtualization Software (e.g., IBM Spectrum Virtualize).
    - **Benefits***:*
        - Simplified storage management.
        - Better utilization of storage resources.
        - Improved data availability and disaster recovery.
6. **Application Virtualization**
    - **Definition:** Technology that allows applications to be abstracted from the underlying operating system, enabling them to run in isolated environments.
    - **Key Technologies:** Citrix XenApp, Microsoft App-V, VMware ThinApp.
    - *Benefits:*
        - Reduced application conflicts.
        - Easier application updates and management.
        - Improved security and isolation.

1.1.1 Key Technologies and Tools

- **Hypervisors:** The core technology enabling hardware virtualization, hypervisors managethe creation and operation of virtual machines.
  - **VMware ESXi:** A widely-used Type 1 hypervisor offering robust performanceand advanced features.
  - **Microsoft Hyper-V:** Integrated with Windows Server, providing an enterprise-grade virtualization platform.
  - **KVM (Kernel-based Virtual Machine):** An open-source hypervisor built intothe Linux kernel, offering flexibility and performance.
  - **Oracle VM VirtualBox:** A popular Type 2 hypervisor for personal and development use.
- **Containers and Orchestration:**
  - **Docker:** A leading platform for developing, shipping, and running containerized applications.
  - **Kubernetes:** An open-source orchestration system for automating thedeployment, scaling, and management of containerized applications.
- **Network Virtualization Technologies:**
  - **VLANs:** Segregate network traffic for improved security and performance.
  - **SDN (Software-Defined Networking):** Decouples the control plane from thedata plane, allowing for programmable network management.
  - **NFV (Network Functions Virtualization):** Virtualizes network servicestraditionally run on dedicated hardware.
- **Storage Virtualization Solutions**:
  - **SAN (Storage Area Network):** Provides high-speed, centralized storage access.
  - **NAS (Network Attached Storage):** Offers file-based storage accessible over anetwork.
  - **Software Solutions:** IBM Spectrum Virtualize, VMware vSAN for storage pooling and management.
- **Application Virtualization Tools:**
  - **Citrix XenApp:** Delivers virtualized applications to users on any device.
  - **Microsoft App-V:** Enables applications to be deployed, updated, and managedcentrally.
  - **VMware ThinApp:** Encapsulates applications in virtual environments to eliminate conflicts and ease deployment.

Benefits of Virtualization

- **Resource Efficiency:** Maximizes the use of hardware resources by running multipleVMs on a single physical machine.
- **Cost Savings:** Reduces the need for physical hardware, leading to lower costs for equipment, maintenance, and power.
- **Scalability:** Easily scale up or down by adding or removing virtual resources.
- **Disaster Recovery:** Simplifies backup and recovery processes since virtual machines

canbe easily replicated and restored.

- **Isolation:** Provides strong isolation between different virtual environments, improvingsecurity and stability.

1.2 Cloud Computing

Cloud computing delivers on-demand access to computing resources (servers, storage, databases, software, etc.) over the internet. Users pay only for the resources they use.

Deployment Models

Cloud Deployment Model functions as a virtual computing environment with a deployment architecture that varies depending on the amount of data you want to store and who has access to the infrastructure.

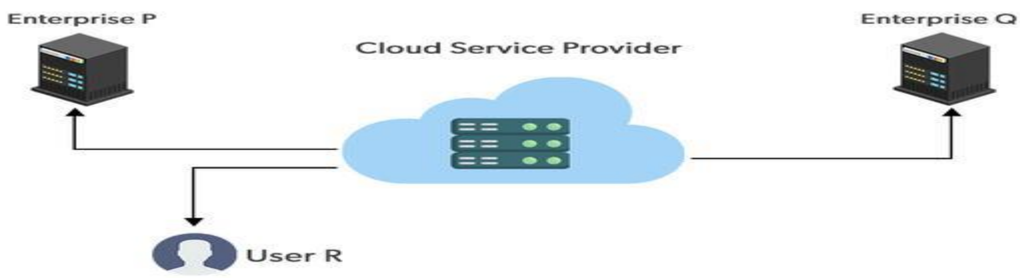
Types of Cloud Computing Deployment Models

The cloud deployment model identifies the specific type of cloud environment based on ownership, scale, and access, as well as the cloud's nature and purpose. The location of the servers you're utilizing and who controls them are defined by a cloud deployment model. It specifies how your cloud infrastructure will look, what you can change, and whether you will be given services or will have to create everything yourself. Relationships between the infrastructure and your users are also defined by cloud deployment types. Different types of cloud computing deployment models are described below.

- Public Cloud
- Private Cloud
- Hybrid Cloud
- Community Cloud
- Multi-Cloud

Public Cloud

The public cloud makes it possible for anybody to access systems and services. The public cloud may be less secure as it is open to everyone. The public cloud is one in which cloud infrastructure services are provided over the internet to the general people or major industry groups. The infrastructure in this cloud model is owned by the entity that delivers the cloud services, not by the consumer. It is a type of cloud hosting that allows customers and users to easily access systems and services. This form of cloud computing is an excellent example of cloud hosting, in which service providers supply services to a variety of customers. In this arrangement, storage backup and retrieval services are given for free, as a subscription, or on a per-user basis. For example, Google App Engine etc.



Public Cloud

Fig 1.2.1

Advantages of the Public Cloud Model

- **Minimal Investment:** Because it is a pay-per-use service, there is no substantial upfront fee, making it excellent for enterprises that require immediate access to resources.
- **No setup cost:** The entire infrastructure is fully subsidized by the cloud service providers, thus there is no need to set up any hardware.
- **Infrastructure Management is not required:** Using the public cloud does not necessitate infrastructure management.
- **No maintenance:** The maintenance work is done by the service provider (not users).
- **Dynamic Scalability:** To fulfill your company's needs, on-demand resources are accessible.

Disadvantages of the Public Cloud Model

- **Less secure:** Public cloud is less secure as resources are public so there is no guarantee of high-level security.
- **Low customization:** It is accessed by many public so it can't be customized according to personal requirements.

Private Cloud

The private cloud deployment model is the exact opposite of the public cloud deployment model. It's a one-on-one environment for a single user (customer). There is no need to share your hardware with anyone else. The distinction between private and public clouds is in how you handle all of the hardware. It is also called the "internal cloud" & it refers to the ability to access systems and services within a given border or organization. The cloud platform is implemented in a cloud-based secure environment that is protected by powerful firewalls and under the supervision of an organization's IT department. The private cloud gives greater flexibility of control over cloud resources.

Private Cloud

Fig 1.2.2

Advantages of the Private Cloud Model

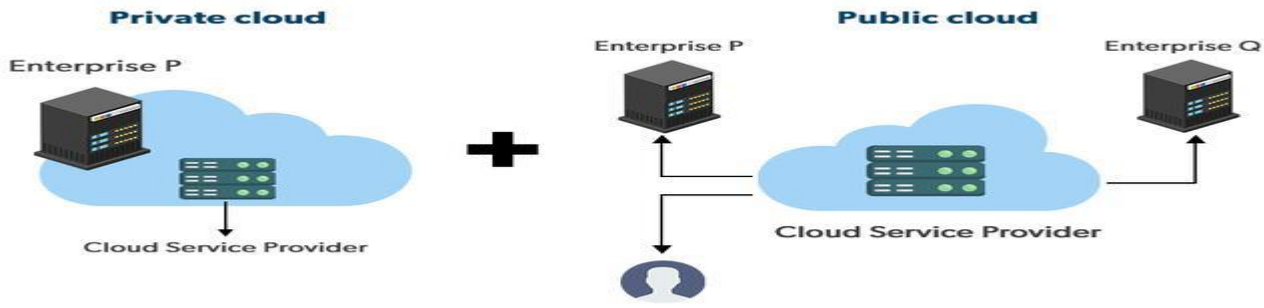
- **Better Control:** You are the sole owner of the property. You gain complete command over service integration, IT operations, policies, and user behavior.
- **Data Security and Privacy:** It's suitable for storing corporate information to which only authorized staff have access. By segmenting resources within the same infrastructure, improved access and security can be achieved.
- **Supports Legacy Systems:** This approach is designed to work with legacy systems that are unable to access the public cloud.
- **Customization:** Unlike a public cloud deployment, a private cloud allows a company to tailor its solution to meet its specific needs.

Disadvantages of the Private Cloud Model

- **Less scalable:** Private clouds are scaled within a certain range as there is less number of clients.
- **Costly:** Private clouds are more costly as they provide personalized facilities.

Hybrid Cloud

By bridging the public and private worlds with a layer of proprietary software, hybrid cloud computing gives the best of both worlds. With a hybrid solution, you may host the app in a safe environment while taking advantage of the public cloud's cost savings. Organizations can move data and applications between different clouds using a combination of two or more cloud deployment methods, depending on their needs.



Hybrid Cloud

Fig 1.2.3

Advantages of the Hybrid Cloud Model

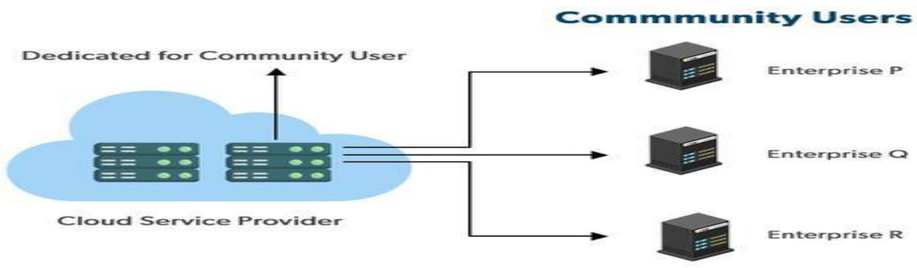
- **Flexibility and control:** Businesses with more flexibility can design personalized solutions that meet their particular needs.
- **Cost:** Because public clouds provide scalability, you'll only be responsible for paying for the extra capacity if you require it.
- **Security:** Because data is properly separated, the chances of data theft by attackers are considerably reduced.

Disadvantages of the Hybrid Cloud Model

- **Difficult to manage:** Hybrid clouds are difficult to manage as it is a combination of both public and private cloud. So, it is complex.
- **Slow data transmission:** Data transmission in the hybrid cloud takes place through the public cloud so latency occurs.

Community Cloud

It allows systems and services to be accessible by a group of organizations. It is a distributed system that is created by integrating the services of different clouds to address the specific needs of a community, industry, or business. The infrastructure of the community could be shared between the organization which has shared concerns or tasks. It is generally managed by a third party or by the combination of one or more organizations in the community.



Community Cloud

Fig 1.2.4

Advantages of the Community Cloud Model

- **Cost Effective:** It is cost-effective because the cloud is shared by multiple organizations or communities.
- **Security:** Community cloud provides better security.
- **Shared resources:** It allows you to share resources, infrastructure, etc. with multiple organizations.
- **Collaboration and data sharing:** It is suitable for both collaboration and data sharing.

Disadvantages of the Community Cloud Model

- **Limited Scalability:** Community cloud is relatively less scalable as many organizations share the same resources according to their collaborative interests.
- **Rigid in customization:** As the data and resources are shared among different organizations according to their mutual interests if an organization wants some changes according to their needs they cannot do so because it will have an impact on other organizations.

Multi-Cloud

We're talking about employing multiple cloud providers at the same time under this paradigm, as the name implies. It's similar to the hybrid cloud deployment approach, which combines public and private cloud resources. Instead of merging private and public clouds, multi-cloud uses many public clouds. Although public cloud providers provide numerous tools to improve the reliability of their services, mishaps still occur. It's quite rare that two distinct clouds would have an incident at the same moment. As a result, multi-cloud deployment improves the high availability of your services even more.

Advantages of the Multi-Cloud Model

- You can mix and match the best features of each cloud provider's services to suit the demands of your apps, workloads, and business by choosing different cloud providers.
- **Reduced Latency:** To reduce latency and improve user experience, you can choose cloud regions and zones that are close to your clients.
- **High availability of service:** It's quite rare that two distinct clouds would have an incident at the same moment. So, the multi-cloud deployment improves the high availability of your services.

Disadvantages of the Multi-Cloud Model

- **Complex:** The combination of many clouds makes the system complex and bottlenecks may occur.
- **Security issue:** Due to the complex structure, there may be loopholes to which a hacker can take advantage hence, makes the data insecure.

Service Models :

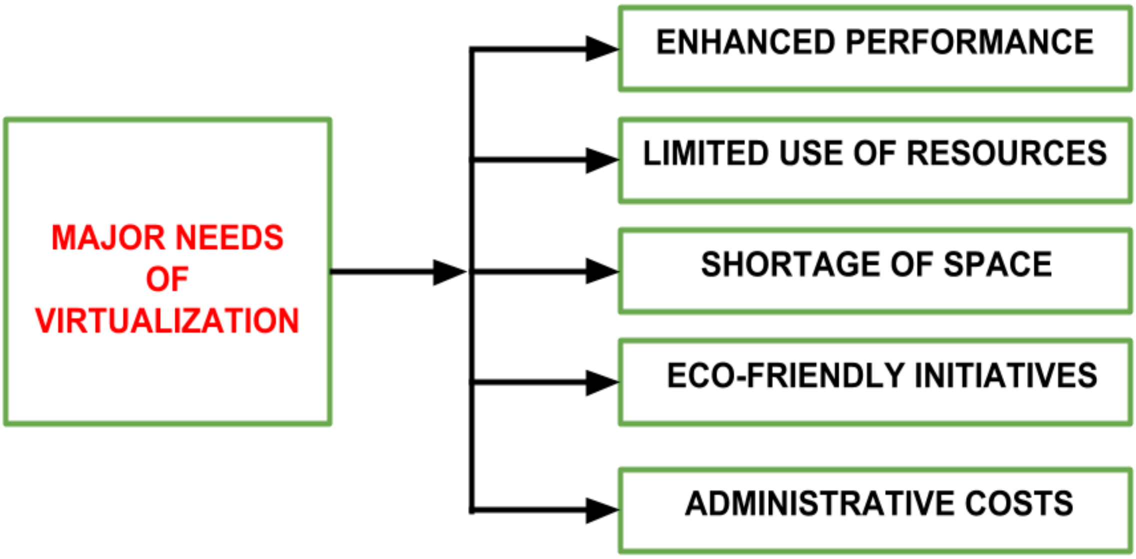
Infrastructure as a Service (IaaS): Provides users with virtualized computing resources like servers, storage, and networking.

Platform as a Service (PaaS): Offers a platform for developing, deploying, and managing applications without managing the underlying infrastructure.

Software as a Service (SaaS): Delivers on-demand access to software applications over the internet. Users access the software through a web browser or API.

1.3 Need for Virtualization:

There are five major needs of virtualization which are described below:



Enhanced Performance

- Modern PCs are powerful enough to handle basic computation requirements and additional tasks.
- These systems have sufficient resources to host virtual machine managers and operate virtual machines with acceptable performance.

Limited Use of Hardware and Software Resources

- Many user systems are underutilized, as they are capable of fulfilling regular computational needs.
- These systems can run continuously without interruptions, offering opportunities for extended usage.
- Virtualization can enable the utilization of idle resources during off-hours, improving IT infrastructure efficiency.

Shortage of Space

- Increasing demand for additional storage or compute capacity pressures organizations to expand data centers.
- Large corporations like Google, Microsoft, and Amazon address these needs by building new data centers, but smaller enterprises may lack the resources to do so.
- Server consolidation through virtualization can optimize existing resources, reducing the need for physical expansion.

Eco-Friendly Initiatives

- Data centers consume significant power for operations and cooling.
- Server consolidation through virtualization reduces the number of physical servers, thereby decreasing power and cooling requirements.
- This aligns with corporate goals to minimize energy expenditures and promote environmental sustainability.

Administrative Costs

- Expanding data center capacity leads to increased administrative tasks, such as hardware monitoring, server setup, updates, and backups.
- These operations are personnel-intensive, leading to higher costs as the number of servers increases.
- Virtualization reduces the required number of physical servers, lowering administrative workloads and costs.

Reduced Costs:

* Consolidate multiple physical servers onto fewer machines, lowering hardware acquisition and maintenance expenses.

* Pay only for the resources VMs use, optimizing resource allocation and eliminating overprovisioning.

Simplified Administration:

* Manage VMs centrally, reducing time spent on individual server configurations.

* Easily provision, migrate, and clone VMs, streamlining IT operations.

Fast Deployment:

Quickly spin up new VMs with pre-configured settings, accelerating application deployment times.

Test and deploy new environments rapidly, improving development agility.

Reduced Infrastructure Footprint:

Consolidate physical servers, minimizing data center space requirements and associated cooling

1.4 Limitations of Virtualization

Cost:

* Upfront investment in virtualization software and potentially new hardware to support VMs.

* Licensing costs for additional operating systems running on VMs.

Complexity:

* Increased management overhead for virtual infrastructure compared to physical machines.

* Requires skilled IT staff to manage VMs, configure virtual networks, and ensure resource allocation.

Performance:

* Virtualization overhead can impact VM performance compared to dedicated physical hardware.

* Overprovisioning VMs on a single physical machine can lead to resource contention and performance bottlenecks.

Security:

* Security vulnerabilities in the virtualization layer can expose all VMs running on the system.

* Managing security across multiple VMs can be complex.

Limited Scalability:

* Scaling resources can be limited by the capacity of the underlying physical hardware.

* Adding physical machines to increase capacity might not be as cost-effective as scaling in a cloud environment.

1.5 HARDWARE VIRTUALIZATION

There are Three types of hardware virtualization:

1.Full virtualization

2.Para Virtulaization

3.Partial Virtualization

Full Virtualization

Full Virtualization is a virtualization technique that simulates an entire physical computer, including its hardware components, to create multiple virtual machines (VMs) that run independently on a single physical host. In this approach, the guest operating system is unaware that it's running in a virtualized environment, as it interacts with virtualized hardware that emulates real hardware.

How Full Virtualization Works?

1. **HypervisorLayer:**
   Full virtualization relies on a hypervisor, also known as a **Virtual Machine Monitor (VMM)**, which sits between the physical hardware and guest operating systems. The hypervisor manages and controls the allocation of physical resources to virtual machines.

2. **HardwareVirtualization:**
   Full virtualization uses hardware-assisted virtualization technologies like **Intel VT-x** or **AMD-V** to enhance performance. These technologies allow the hypervisor to run guest OSes directly on the physical CPU without significant performance overhead.

3. **Isolation:**
   VMs created through full virtualization are completely isolated from each other. Each VM runs its own instance of the guest operating system, which cannot interfere with other VMs.

4. **Examples:**
   Popular hypervisors for full virtualization include **VMware vSphere/ESXi**, **Microsoft Hyper-V**, and **Oracle VirtualBox**.

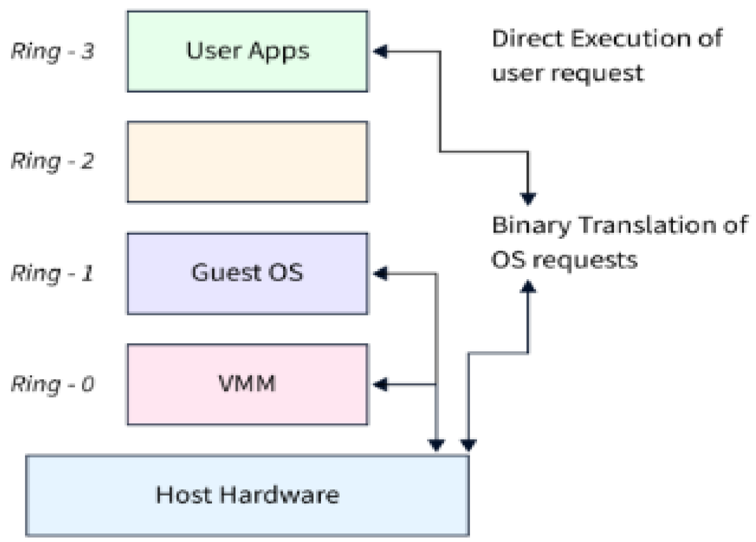


Fig 1.5.1

Full virtualization is well-suited for scenarios where you need to run different operating systems on the same physical server, such as hosting multiple Windows and Linux VMs on a single machine. It offers excellent isolation and security between VMs, making it suitable for scenarios like cloud computing and data center environments.

Para Virtualization

Para Virtualization, on the other hand, takes a slightly different approach to virtualization. It involves modifying the guest operating systems to be aware of the virtualized environment. Unlike full virtualization, where the guest OS runs unmodified, paravirtualization requires guest OSes to use a specific set of APIs to interact with the virtualization layer.

How Para Virtualization Works?

1. **HypervisorLayer:**
   Similar to full virtualization, paravirtualization also employs a hypervisor, but here, the guest operating systems are aware of it. The hypervisor provides a set of APIs that guest OSes must use to communicate with the underlying hardware.

2. **GuestOSModifications:**
   Guest operating systems must be modified to replace certain hardware-related instructions with hypercalls, which are calls to the hypervisor. These hypercalls allow the guest OS to request services from the hypervisor, such as memory management or CPU scheduling.

3. **PerformanceBenefits:**
   Since para virtualization avoids the overhead of emulating complete hardware, it often offers better performance than full virtualization. Guest OSes can communicate more directly with the hypervisor, resulting in improved efficiency.

4. **Examples:**
   Xen is a widely-used hypervisor that supports para virtualization. It is known for its performance and scalability in virtualized environments.

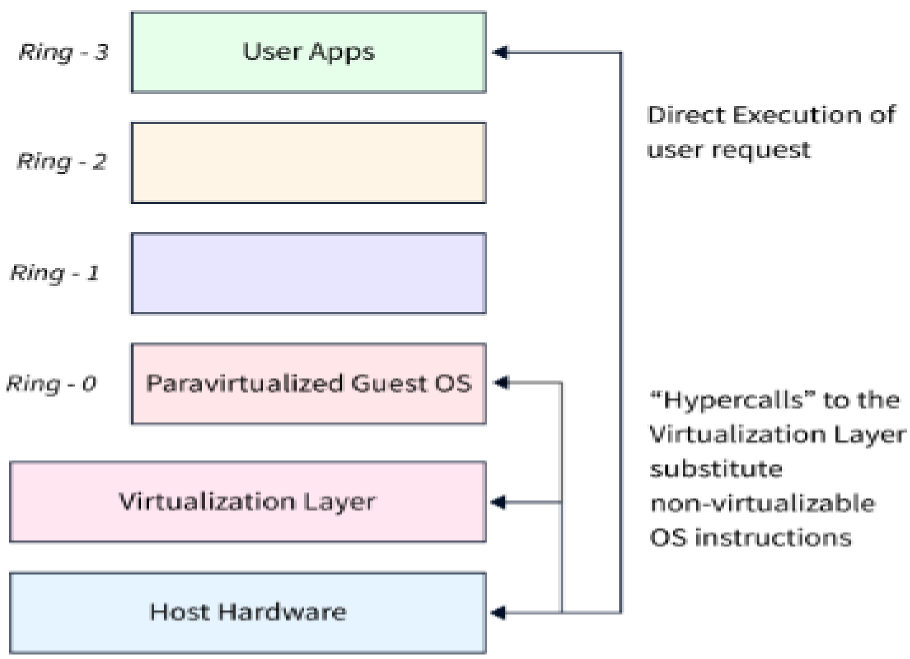


Fig 1.5.2

Full Virtualization Vs Para Virtualization

Let's summarize the key differences between full virtualization and para virtualization in a tabular format:

Aspect	**Full Virtualization**	**Para Virtualization**
Guest OS Modification	Not required; runs unmodified	Requires modifications to use hypercalls
Performance	Slightly lower due to emulation	Better performance due to direct interaction
Isolation	Strong isolation between VMs	Isolation with awareness of other VMs
Guest OS Flexibility	Supports various OS types	Works best with compatible OSes
Hardware Compatibility	Compatible with most hardware	Requires hardware support for para virtualization
Hypervisor Layer	Hypervisor manages virtual hardware independently	Hypervisor provides APIs for communication
Interaction with Hardware	Emulates complete hardware	Uses hypercalls to request hardware services
Examples	VMware, Hyper-V, VirtualBox	Xen, KVM, QEMU
Resource Overhead	Slightly higher resource overhead	Lower resource overhead for the hypervisor
Use Cases	Mixed OS environments, strong isolation needed	Performance-critical applications, homogeneous OS environment

Partial Virtualization

Definition:
Partial virtualization involves virtualizing only certain parts of the underlying hardware, such as CPU or memory, while other components are accessed directly by the guest operating system.

Key Features:

1. **Direct Hardware Access**: The guest operating system directly accesses some hardware components, bypassing the hypervisor.
2. **Modification Required**: The guest operating system must be modified to work in this environment.
3. **Performance**: Offers better performance than full virtualization due to reduced overhead.

Advantages:

- Efficient use of system resources.
- Lower overhead compared to full virtualization.

Disadvantages:

- Requires modification of the guest OS, making it less flexible.
- Limited isolation compared to full virtualization.

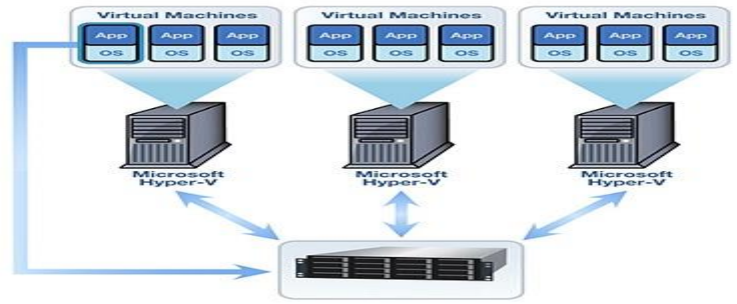
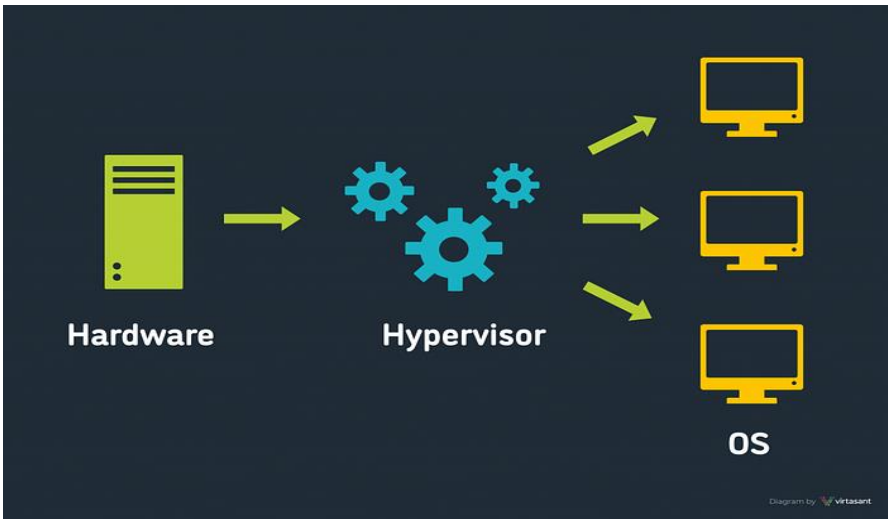


Fig 1.5.3

1.6 Hypervisor

What is a Hypervisor?

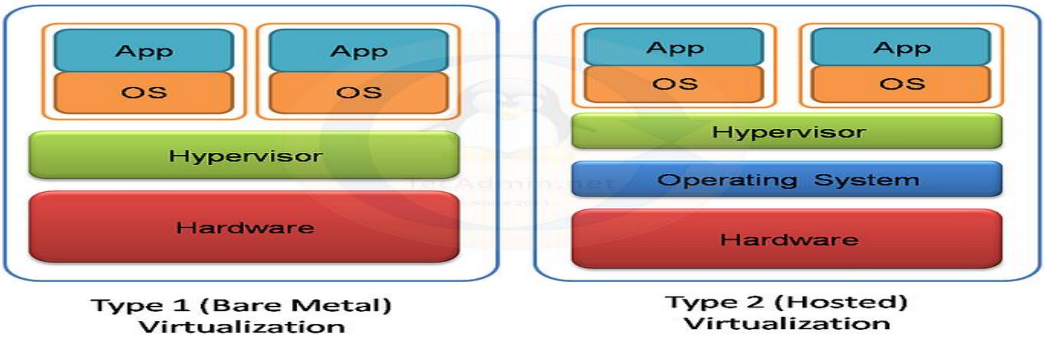
A hypervisor, also known as a virtual machine monitor (VMM), is a software or firmware layer that enables multiple operating systems, known as guest operating systems, to run concurrently on a single physical host. The hypervisor abstracts and partitions the underlying hardware resources, such as CPU, memory, storage, and networking, to create isolated virtual environments for each guest operating system.



Hypervisor

Fig 1.6.1

Type-1 Hypervisor vs Type-2 Hypervisor:



Type — 1 vs Type 2 Virtualization

Fig 1.6.2

Type-1 Hypervisor (Bare Metal Hypervisor):

- Type-1 hypervisors run directly on the physical hardware without the need for a host operating system.
- They have direct access to the underlying hardware resources, offering better performance and efficiency compared to Type-2 hypervisors.
- Type-1 hypervisors are commonly used in enterprise data centers and cloud environments.
- Examples include VMware vSphere/ESXi, Microsoft Hyper-V, and KVM (Kernel-based Virtual Machine).

Type-2 Hypervisor (Hosted Hypervisor):

- Type-2 hypervisors run on top of a host operating system, which in turn runs on the physical hardware.

- They rely on the host operating system for managing hardware resources and providing device drivers.
- Type-2 hypervisors are typically used for desktop or development environments.
- Examples include Oracle VirtualBox, VMware Workstation, and Parallels Desktop.

SUMMARY OF DIFFERENCES: TYPE 1 VS. TYPE 2 HYPERVISORS

	Type 1 hypervisor	**Type 2 hypervisor**
Also known as	Bare metal hypervisor.	Hosted hypervisor.
Runs on	Underlying physical host machine hardware.	Underlying operating system (host OS).
Best suited for	Large, resource-intensive, or fixed-use workloads.	Desktop and development environments.
Can it negotiate dedicated resources?	Yes.	No.
Knowledge required	System administrator-level knowledge.	Basic user knowledge.
Examples	VMware ESXi, Microsoft Hyper-V, KVM.	Oracle VM VirtualBox, VMware Workstation, Microsoft Virtual PC.

Pros and Cons of Hypervisor

Pros:

1. Efficient resource utilization and cost savings.
2. Supports multiple OSs on one machine.
3. Isolation prevents VM interference.
4. Scalable and flexible for workload changes.
5. Backup, recovery, and live migration for high availability.
6. Ideal for testing and cloud integration.

Cons:

1. Performance overhead and latency.
2. Single point of failure if hypervisor crashes.
3. Complex management and troubleshooting.
4. Security risks like hypervisor vulnerabilities.
5. Resource contention among VMs.
6. High initial setup costs.

ADVANTAGES OF HARDWARE VIRTUALIZATION

- Lower Cost: Because of server consolidation, the cost decreases; now, multiple OS can exist together in a single hardware. This minimizes the quantity of rack space, reduces the number of servers, and eventually drops the power consumption.
- Efficient resource utilization: Physical resources can be shared among virtual machines. Another virtual machine can use the unused resources allocated by one virtual machine in case of any need.
- Increase IT flexibility: The quick development of hardware resources became possible using virtualization, and the resources can be managed consistently also.
- Advanced Hardware Virtualization features: With the advancement of modern hypervisors, highly complex operations maximize the abstraction of hardware & ensure maximum uptime. This technique helps to migrate an ongoing virtual machine from one host to another host dynamically.